

STATISTICS

PROOFS FOR REGRESSION ANALYSIS

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Proofs for Regression Analysis

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Regression Line

Population Parameter	Sample statistic
β_0	b_0
β_1	b_1
$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$	$\hat{Y}_i = b_0 + b_1 X_i$

Here;

- Y_i : the observation point the dependent variable and it is random variable
- X_i : the observation point the independent variable and it is not random variable
- β_0 : the parameter of the population and it is not random
- β_1 : the parameter of the population and it is not random
- b_0 : the statistic of the sample and it is random
- b_1 : the statistic of the sample and it is random
- ε_i : error term and it is random. Here, we assume that ε_i 's are independent and

$$\varepsilon_i \sim N(0, \sigma^2)$$

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Normal equations of the line are:

$$b_1 = \frac{\sum_{i=1}^n X_i Y_i - \frac{(\sum_{i=1}^n X_i)(\sum_{i=1}^n Y_i)}{n}}{\sum_{i=1}^n X_i^2 - \frac{(\sum_{i=1}^n X_i)^2}{n}} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$b_0 = \bar{Y} - b_1 \bar{X}$$

Theorem 1: $E[b_1] = \beta_1$ (b_1 is an unbiased estimator of β_1)

Proof 1:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})(\beta_0 + \beta_1 X_i + \varepsilon_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})(\beta_1 X_i + \varepsilon_i)}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\sum_{i=1}^n (X_i - \bar{X})X_i = \sum_{i=1}^n (X_i - \bar{X})^2 \quad \Rightarrow \quad b_1 = \beta_1 + \frac{\sum_{i=1}^n ((X_i - \bar{X})\varepsilon_i)}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad E[\varepsilon_i] = 0$$

$$\Rightarrow E[b_1] = E\left[\beta_1 + \frac{\sum_{i=1}^n ((X_i - \bar{X})\varepsilon_i)}{\sum_{i=1}^n (X_i - \bar{X})^2}\right] = E[\beta_1] + \frac{\sum_{i=1}^n [(X_i - \bar{X})E[\varepsilon_i]]}{\sum_{i=1}^n (X_i - \bar{X})^2} = E[\beta_1] = \beta_1$$

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Theorem 2: $E[b_2] = \beta_2$ (b_2 is an unbiased estimator of β_2)

Proof 2:

$$b_0 = \bar{Y} - b_1 \bar{X}$$

$$E[b_1] = \beta_1$$

$$\Rightarrow E[b_0] = E[\bar{Y} - b_1 \bar{X}] = E[\bar{Y}] - \bar{X} E[b_1] = \underbrace{\beta_0 + \beta_1 \bar{X} - \bar{X} \beta_1}_{\beta_0} = \beta_0$$

$$E[\bar{Y}] = E\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n} \sum_{i=1}^n E[Y_i] = \frac{1}{n} \sum_{i=1}^n E[\beta_0 + \beta_1 X_i + \varepsilon_i] = \beta_0 + \beta_1 \bar{X}$$

$E[\varepsilon_i] = 0$

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Theorem 3:
$$Var[b_1] = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sigma^2}{\sum_{i=1}^n X_i^2 - \frac{(\sum_{i=1}^n X_i)^2}{n}}$$

Proof 3:

In proof 1, we obtained that
$$b_1 = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X})\varepsilon_i}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

➡
$$Var[b_1] = Var\left[\beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X})\varepsilon_i}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]$$

ε_i 's are independent

$$= \underbrace{Var[\beta_1]}_0 + \frac{Var[\sum_{i=1}^n (X_i - \bar{X})\varepsilon_i]}{(\sum_{i=1}^n (X_i - \bar{X})^2)^2}$$

$$= \frac{\sum_{i=1}^n (X_i - \bar{X})^2 Var(\varepsilon_i)}{(\sum_{i=1}^n (X_i - \bar{X})^2)^2} = \frac{Var[\varepsilon_i] = \sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

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Theorem 4: $Var[b_0] = \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right) \sigma^2 = \frac{\sum_{i=1}^n X_i^2}{n \sum_{i=1}^n X_i^2 - (\sum_{i=1}^n X_i)^2} \sigma^2$

Proof 4:

$$b_0 = \bar{Y} - b_1 \bar{X}$$

$$\Rightarrow Var[b_0] = Var[\bar{Y} - b_1 \bar{X}] = \underbrace{Var[\bar{Y}]} + (-1)^2 \bar{X}^2 \underbrace{Var[b_1]} = \frac{\sigma^2}{n} + \frac{\bar{X}^2 \sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$Var[b_1] = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$Var[\bar{Y}] = Var\left[\frac{1}{n} \sum_{i=1}^n Y_i\right] = \frac{1}{n^2} \sum_{i=1}^n \underbrace{Var[Y_i]} = \frac{1}{n^2} n \sigma^2 = \frac{\sigma^2}{n}$$

$$Var[Y_i] = Var[\beta_0 + \beta_1 X_i + \varepsilon_i] = Var[\varepsilon_i] = \sigma^2$$